MILITARY SPECIFICATION

BOAT SYSTEM, UTILITY, NOMINAL, 20 FOOT REINFORCED PLASTIC WITH 85 HORSEPOWER OUTBOARD ENGINE AND TRAILER

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type and size reinforced plastic commercial utility boat system, with 85 horsepower (hp), outboard engine trailer, and with all other necessary components and accessories to provide a complete, safe, usable boat for use in placing oil spill barriers.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU-T-81 - Tags, Shipping and Stock.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-636 - Boxes, Shipping, Fiberboard.

MILITARY

MIL-C-104 - Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116 - Preservation-Packaging Methods of.
MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-C-3774 - Crates, Wood; Open, 12,000- and 16,000-Pound Capacity.

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

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MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.

(Copies of specifications, standards, and publications required by contractor's in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

US COAST GUARD REGULATIONS (USCG)

Title 46, Code of Federal Regulation, Parts 24 and 25.

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

AMERICAN BOAT AND YACHT COUNCIL (ABYC)

Safety Standards for Small Craft.

(Application for copies should be addressed to the American Boat and Yacht Council, Inc., 15 East 26th Street, Room 1603, New York, NY 10010.)

BOATING INDUSTRY ASSOCIATIONS (BIA)

BIA Certification Handbook.
Trailer Manufacturer's Association.

(Application for copies of BIA Certification Handbook should be addressed to the Boating Industry Association, 401 N. Michigan Avenue, Chicago, IL 60601.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., Tariff Order Section, 1616 P Street, N. W. Washington, DC 20036.)

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TIRE AND RIM ASSOCIATION, INC.

T&RA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Description. The utility boat system shall consist of a manufacturer's standard, BIA certified, nominal 20 foot, reinforced plastic, boat with trailer and 85 hp outboard engine. The boat shall be a multiple V hull type with chine and without decks. All accessories and safety equipment required for safe, efficient operation shall be furnished as specified herein. The engine shall be a commercial, two-cycle, long shaft, electric starting, outboard type. The trailer shall be a commercial boat trailer having a minimum rated capacity of 3,000 pound (lb). The trailer shall be certified by the Trailer Manufacturer's Assosiation, (TMA).

3.2 First article. When specified (see 6.2), the contractor shall furnish one complete boat system for first article inspection and approval (see 4.3 and 6.4).

3.3 Material. All material shall be new and unused. All material shall be as specified herein. Material not specified shall be of the same quality used for the intended purpose in commercial practice. Materials used shall be free from defects which would adversely affect the performance or maintainability of the individual component or of the overall system.

3.3.1 Reinforcing material. Reinforcing material shall be either fiberglass cloth or a suitable synthetic fiber substitute. Synthetic fiber laminate shall have strength characteristics equivalent or superior to the fiberglass laminate it replaces.

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3.3.2 Polyester resin. Polyester resin shall be the manufacturer's standard fire retardant type, for use with the reinforcing material and the plastic foam specified.

3.3.3 Hull buoyancy material (plastic foam). Buoyancy material shall be closed cell rigid polyurethane foam of suitable density and in sufficient quantity to support the flooded boat, plus one battery weighing 40 lb, fully loaded fuel tanks, one outboard engine weighing 300 lb, and an additional 1,000 lb. The buoyancy material shall be foamed in place and chemically crosslinked to the hull shells.

3.3.4 Metal. Metal used in the utility boat system shall have sufficient yield strength and section modulus to withstand imposed loadings. Additionally, all exposed metal surfaces shall be provided with protection from a salt water corrosive environment (see 3.12).

3.4 Standard commercial product. The utility boat system shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product shall be included in the utility boat system being furnished. Standard commercial product is a product which has been or will be sold on the commercial market through advertisements or manufacturer's catalogs or brochures, and represents the latest production model(s).

3.5 Boat design. The boat shall be constructed of an outer and inner shell of fiberglass, or suitable synthetic fiber, reinforced polyester resin with buoyancy material between the two shells. The buoyancy material shall be chemically crosslinked to the shell material to preclude possible delamination. The method used to obtain crosslinking shall be the manufacturer's standard method. The outer shell shall be a minimum of 3/16-inch thick at the chine and 1/4-inch thick at the keel. Reinforcing material shall contribute adequate strength to the structure to resist static and dynamic loads. The hull shall have no structural members constructed of wooden material. The boat shall have a transom compatible with a long shaft outboard engine. The boat shall be equipped with a self-draining outboard mounting well. The well shall have provisions for installation of a shroud, if required (see 6.2), to prevent excessive water entry. The boat shall have a fitting for drainage of accumulated water, either on land or when cruising. The boat shall be rated for an outboard engine capacity of no less than 135 hp. Construction shall be strong and durable, suitable for salt water use, and shall conform to ABYC, USCG and BIA standards. The boat shall have good seakeeping qualities for a boat of this size at speeds up to 40 knots.

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3.5.1 Overall boat specification.

Length	19 - 21 ft
Beam	6-1/2 - 8 ft
Freeboard with 1500 lb load	
Amid ship	16^{+0}_{-2} inch
Transom height	Nominal 20 inch suitable for longshaft outboard
Weight	Not to exceed 1,900 lb
Finish (exterior and interior)	Impact resistant impregnated gel coat

3.5.2 Interior configuration.

3.5.2.1 Console. The steering/engine control console shall be positioned 1/3 to 1/2 of the length of the boat behind the bow. The console shall be constructed of reinforced plastic similar to the hull material. The console size shall allow for ease of personnel passage between the bow and stern of the boat. Unless otherwise specified, (see 6.2) the console shall be positioned on the starboard side of the boat. The console shall be equipped with a rigid windscreen of console width. The windscreen shall be a minimum of 18 inches high and shall be provided with adequate framing and bracing to insure safe operation. Windscreen material shall meet ABYC recommendations.

3.5.2.2 Seat. One swivel type contoured seat made of fiberglass reinforced plastic shall be positioned on a cast marine aluminum pedestal and bolted in place, directly astern of the console. The bucket seat shall be cushioned and covered with waterproof material and shall have no arms. The seat is to be positioned so as to allow operator to both stand and sit behind the steering wheel.

3.5.2.3 Stern section. The stern section shall be left undecked so as to allow ventilation of, and access to, portable type engine fuel tanks where utilized and to electrical connections, and battery box.

3.5.2.4 Railing. Approved marine railing shall be installed on the boat. The side railing shall be manufacturer's standard length 4-inch railing and shall be installed on both sides of the boat. The bow railing shall be the manufacturer's standard bow railing.

3.5.2.5 Floor covering. The floor of the interior shall be equipped with a non-skid surface.

3.5.3 Boat cover. A full length, form fitted, easily installed cover of cotton duck shall be furnished with each boat to provide protection during its storage.

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3.5.4 Towing accessories. The boat shall be equipped with a towing device located at the turning center of the boat. The towing device shall be designed, as a minimum, to withstand six times the rated thrust of the boat-engine-propeller combination without failure, deformation or cracking of the towing device or the boat. The design of the towing device shall also prevent interference between the engine and the article being towed.

3.5.5 Mooring attachments. The boat shall be equipped with at least five mooring cleats, one located in each corner, and one at the prow. A bow eye, used for boat mooring and towing shall be installed on the forward side of the stem.

3.5.6 Hoisting fittings. The boat shall have three hoisting fittings inside the boat, two of which shall be located at the transom, port and starboard, and one shall be located forward in the stem. Each hoist fitting, and its mounting, shall be rated for loads through 2,000 lb.

3.5.7 Safety equipment. Each utility boat system shall be equipped with USCG approved safety equipment required for a boat of the size and type specified herein. The equipment shall be mounted along the boat inside walls using appropriate mounting brackets. All equipment shall satisfy subchapter C, 46 Code of Federal Regulation, parts 24 & 25. Equipment shall include but not be limited to the following:

2 each - 5 foot long wooden paddles.
1 each - 8 foot aluminum boat hook.
1 each - 5 lb dry chemical Purple-K-Powder fire extinguisher.
1 each - First aid kit.
1 each - Emergency flare kit.
1 each - 15 lb Danforth anchor.
1 each - 100-foot length of 1/2-inch nylon mooring line with stainless steel thimble and eye splice at one end.
6 each - Life preservers, jacket type, fibrous glass filled.
4 each - Fastened in place rubber foot step pads.
1 each - Compressed air fog horn.
2 each - Vinyl, foam filled, 24-inch long fenders.

3.5.8 Electrical accessories. The boat shall be furnished with running lights conforming to USCG regulations and requirements (subchapter C 46 CFR, parts 24 and 25) for International waters, complete with wiring for 12 volts, direct current (dc), and waterproof switch mounted on the control console. One, variable direction, portable, switch operated, 12 volt, sealed beam spotlight of not less than 35,000 candle power, shall be installed on the control console. The spotlight shall have a 6-foot stretch cord with waterproof plug connected into a waterproof outlet. A heavy duty marine type battery designed for starting the furnished engine shall be provided in a dry

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charged condition, in a plastic battery box with cover. The battery box shall be secured to the boat floor with a removable strap in a convenient location in the stern of the boat.

3.5.9 Rubber bumpers. A rubber D-type bumper shall be installed at the leading vertical edge of the hull. This fendering system shall be installed on the full perimeter of the boat except for the transom area. An additional bumper shall be installed, at the waterline, on the side of the hull, if this is not the leading vertical edge. The D-type fender material shall extend a minimum of 3 inches from the hull and shall have a minimum vertical dimension when installed, of 3 inches.

3.6 Engine design. The engine shall be a commercially available, two-cycle, electric starting, water-cooled outboard engine rated at 85 hp. The engine shall be installed with all attachments and connections assembled and functioning. The motor leg shall be streamlined to minimize drag and prevent water from entering the boat over the transom.

3.6.1 Performance. The outboard motor shall perform as specified herein when using the gasoline and oil specified in 3.6.17. Under all test conditions specified herein, the outboard motor shall start, run, and stop without loosening, failure, or permanent deformation or excessive wear of any components in ambient temperatures between 32° Fahrenheit (F), and 120°F. When the engine propeller combination is operated at the speed corresponding to the manufacturer's continuous duty rating, the static thrust shall be 800 lb minimum. The supplier shall submit to the contracting officer a maximum recommended operating speed in revolutions per minute (rpm) for the outboard motor. If cavitation is experienced during the tests specified herein, the engine speed shall either drop off or the engine shall be able to sustain the overspeed without failure.

3.6.2 Controls. A single, manually operated control lever, completely mechanical, which regulates engine throttling, gear shifting and provides a receptable for the ignition key shall be provided. The control mechanism shall be equipped with appropriate length control cables and connectors, fastened along the interior side of the boat, to link the mechanism to the furnished outboard engine. The control lever shall be capable of operating the engine through a complete engine cycle (neutral to maximum forward, to neutral, to maximum reverse, to neutral) repeatedly without loosening, permanent deformation, or failure of any component. All controls shall be designed for operation by personnel wearing cold weather gloves. All controls and their positions shall be clearly identified (see 3.15.3).

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3.6.3 Choke. A remote control manually operated choke switch with connections to the engine shall be provided.

3.6.4 Steering. A 15-inch (minimum) diameter steering wheel operating a rack and pinion steering with a push-pull cable to the transom shall be provided. The cable shall be fastened along the interior of the boat side. Steering brackets appropriate to the supplied outboard engine shall be furnished to connect the engine to the steering cable.

3.6.5 Tachometer. An appropriate (0-8000 rpm) heavy-duty tachometer and connections shall be installed to monitor engine speed and to prevent excessive engine loading. The tachometer shall be provided with a red line at the maximum rated rpm.

3.6.6 Cooling. When specified (see 6.2), an appropriate water temperature or water pressure gage and connections shall be installed to monitor cooling water when the engine is used under heavy loading.

3.6.7 Ammeter. When specified (see 6.2), an appropriate ammeter gage and connections shall be installed to monitor the electrical charging system.

3.6.8 Engine log meter. When specified (see 6.2), an appropriate engine operating log meter and connections shall be installed which record operations usage time in hours. An engine log meter having a totalizing mechanism of not less than 9,000 hours shall be furnished. The meter shall accurately record the total engine operating time and provide trouble-free performance under the operating conditions indicated in this specification.

3.6.9 Fuel system. A means for manual priming of the fuel system shall be furnished.

3.6.9.1 Carburetor. The carburetor(s) shall be the Venturi type with fixed high and low speed jets.

3.6.9.2 Fuel tanks. Manufacturer's standard fuel tanks shall be provided. Tanks and tank installation shall meet ABYC, USCG and BIA standards. Minimum overall fuel capacity shall be 18 gallons. Fuel level indication in the tank or tanks shall be provided. If multiple tanks are utilized they shall be rigged to feed the engine consecutively. Filler caps shall be attached to the tanks by detachable chain. If portable fuel tanks are utilized they shall be securely positioned in place, utilizing removable type floor harness straps along the transom of the boat.

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3.6.9.2.1 Fuel lines. Fuel lines shall be installed to prevent chafing by moving parts or vibration, shall be strain free, and shall contain no loops or rises which could cause vapor lock. The fuel supply line shall contain an engine manufacturer's standard fuel filter. The fuel line shall be not less than 6 feet long and shall have a quick-disconnect fitting for connecting to the engine fuel inlet. The fuel line shall be gasoline and oil resistant and shall not deteriorate due to aging, sunlight, heat, cold, or salt water. An automatic valve shall be provided in the quick-disconnect fitting to close off the fuel line from the fuel tank when the line is disconnected from the engine.

3.6.10 Electrical system. Engine shall have a generator or alternator to recharge the furnished battery, and power all electrical accessories installed on the boat. The ignition system shall be of the manufacturer's standard with protection from contamination and waterproof coils. Ignition points, where applicable shall be of corrosion-resistant material and shall be adjustable without using special tools. Ignition timing marks shall be provided. All wiring shall be installed to prevent straining or chafing of components during operation. Engine shall be provided with an electric starting mechanism of the manufacturer's standard. Provisions shall be made for manual starting. A heavy-duty marine battery shall be furnished (see 3.5.8).

3.6.11 Exhaust system. An engine manufacturer's standard underwater exhaust system shall be provided.

3.6.12 Cooling system. A positive circulating water cooling system shall be provided. The system shall be self-draining in the operating position when the outboard motor is not running.

3.6.13 Propeller. The propeller shall have three or four blades and shall be fabricated of manganese bronze, or heat-treated, semipermanent or permanent mold, aluminum-alloy casting as specified (see 6.2). The size (diameter and pitch) of the propeller shall be selected for compatibility with 3.6.1 and shall be die stamped on the hub. A slip-type clutch, integral with the propeller, shall be furnished which will prevent damage to the outboard motor when the propeller strikes an underwater object.

3.6.14 Housings. Outboard motor components such as engine, gears, and pump shall be enclosed in housings of corrosion-resistant material. The housings shall protect the components from damage under normal operating, handling, and shipping conditions. The housings shall afford protection to operating personnel from the hazards of moving parts and parts subject to high temperatures. The housings shall permit access, without the use of special tools, to spark plugs and carburetor for servicing and repair.

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3.6.15 Mounting bracket. The transom mounting bracket shall provide secure attachment to transoms for the minimum range of slopes from 0° to 15° from the vertical without impairment of steering or free tilting in forward and neutral operation. The bracket shall hold the motor while operating at full power in forward and reverse gears. The motor shall not tilt while operating in reverse gear at full power.

3.6.16 Tilt handle. A tilt handle, with sufficient strength and anchor to lift the complete outboard motor, shall be located near the top rear corner of the engine cover to permit tilting the outboard motor forward when mounted on a boat transom.

3.6.17 Fuel mixture. The outboard motor shall be tested with a fuel mixture containing the following components in the ratio specified by the supplier for normal operation.

3.6.17.1 Gasoline. The engine furnished shall produce the required performance on unleaded or low-lead content gasoline. The content of antiknock compounds (metallic lead) for low-lead gasoline shall not exceed 0.5 gram per gallon. Unless otherwise specified (see 6.2), regular grade (92 octane class) gasoline shall be used.

3.6.17.2 Oil. Oil shall conform to the engine manufacturer's recommendations.

3.6.18 Lubricants. The outboard motor shall be assembled, run-in (broken-in), tested, and delivered with the engine manufacturer's recommended lubricants.

3.7 Trailer design. The utility boat system shall include a commercial boat trailer having a rated capacity of 3,000 lb. The trailer shall be certified by the TMA. The trailer shall include all necessary electrical and mechanical equipment for connection to an available Navy pick-up truck or other towing vehicle. Additionally all necessary equipment for loading, securing, and unloading the 20-foot utility boat and accessories shall be provided.

3.7.1 Performance. Vehicle performance as specified in 3.7.1.1 through 3.7.1.5 shall be demonstrated by the trailer, when fully equipped with all attachments and accessories, and with the 20-foot utility boat, engine, and accessories loaded aboard. Trailer shall be completely serviced and then towed, by a towing vehicle, over improved and unimproved hard, uneven terrain. Vehicle shall show no evidence of failure or malfunction of trailer, boat, engine, components, attachment, and accessories.

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3.7.1.1 Towing speed and trailing ability. When transporting the utility boat, engine, and all accessories, while being towed over dry, level, smooth, improved, and prepared hard-surface roads at sustained speeds to 50 miles per hour (mph), and over dry unimproved and gravel roads at speeds to 10 mph, the trailer shall follow the towing vehicle without swerving from side to side over 3 inches to each side of the path of the towing vehicle when it is moving in a straight line.

3.7.1.2 Manual maneuvering. The trailer shall have provisions for short distance, manual maneuvering by two men.

3.7.1.3 Storage and hookup. The trailer shall have provisions for horizontal storage, when not connected to a tow vehicle. When stored fully loaded, the trailer shall be capable of being attached to a trailer hitch equipped pick-up truck, by one man without requiring any manual lifting.

3.7.1.4 Onloading and offloading. With the trailer attached to the truck, the utlity boat, engine, and accessories shall be capable of being unfastened from the trailer and offloaded into the water, utilizing the attached electric winch or the manual override feature. Onloading of the boat onto the trailer shall also be performed utilizing either the electric winch or the manual override feature. The trailer bed, tilt-down feature shall be utilized in onloading and offloading. All devices shall perform with no evidence of failure or malfunction.

3.7.1.5 Turning ability. When coupled to a towing vehicle operating in its minimum turning circle, the trailer shall follow without cramping or side slipping. The trailer shall be capable of assuming an angle of 60° with the towing vehicle without damage to either, or interference between vehicles.

3.7.2 Trailer components.

3.7.2.1 Suspension. Trailer shall be equipped with a suspension system having a rated capacity at least equal to the load imposed on each member, measured at the ground, with vehicle loaded as specified in 3.7. When suspension capacity is rated at the spring pads, unsprung weight shall be deducted.

3.7.2.2 Axle. Axle rating shall be at least equal to the load imposed on the axle, measured at the ground with vehicle loaded as specified in 3.7.1.

3.7.2.3 Wheels, rims, tires, and tubes. Vehicle wheels shall be manufacturer's commercial standard. Rims, and tire ratings shall conform to T&RA recommendations, for the type and size of tires furnished. Tire and rim sizes shall be the same for all wheels on each vehicle.

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3.7.2.4 Tires. Tires shall be tube- or tubeless-type with highway tread. Tires shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel at the ground, with vehicle loaded as specified in 3.7.1. Tires shall be of not less than 100 level quality. Manufacturer's current size of commercial tires conforming to specified requirements may be furnished.

3.7.2.4.1 Inner tubes. When tube-type tires are furnished, inner tubes shall be of heavy-duty type, and shall be of proper size for tires furnished.

3.7.2.5 Bearing lubrication fittings. External bearing lubrication fittings shall be provided for each hub.

3.7.2.6 Spare wheel assembly. Unless otherwise specified (see 6.2), a spare wheel or rim shall be mounted on a carrier and installed in a readily accessible location. Carrier shall permit ready removal or installation of spare assembly.

3.7.2.6.1 Spare tire. Unless otherwise specified (see 6.2), an inflated spare tire mounted on the spare wheel or rim shall be furnished. The spare tire shall be of the same size, tread design, and ply rating as the tires in use on the vehicle.

3.7.2.7 Brakes. The trailer shall be provided with manufacturer's standard hydraulic surge brakes on each wheel. The brake system shall be provided with an automatic breakaway actuation device.

3.7.2.8 Lighting and electrical systems. Trailer lighting and electrical systems shall comply with the ABYC and BIA standards.

3.7.2.8.1 Lighting. The electrical lighting system shall be 12 volt, and shall conform to ABYC and BIA standards. Lights and reflectors shall not be mounted on rubrails or vehicle bumpers.

3.7.2.8.2 Lighting cable. Trailer lighting connectors, coupling sockets, and cable shall conform to ABYC and BIA standards. Coupling socket on trailer shall be protected with cover assembly. Lighting cable shall be of sufficient length to connect the trailer to the towing vehicle and to allow adequate clearance for turning. Provisions shall be made to secure connector socket and cable to drawbar or cargo body, when not in use.

3.7.2.8.3 Turn signals. Turn signal lamps conforming to ABYC and BIA standards shall be furnished.

3.7.2.9 Landing leg. Trailer shall have a retractable landing leg with caster type landing wheel, with cranking handle (rigid or swivel type) permanently installed. The landing leg and related parts shall withstand, without permanent deformation, the combined static and dynamic

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forces due to the proportionate weight of the gross weight, and impacts during coupling and uncoupling operations. Landing leg shall operate freely, and shall be protected from malfuctioning by foreign matter. When placed into travel position, the landing leg shall remain positively locked to provide ground clearance in the course of normal towing operations.

3.7.2.10 Coupling device and towbar. Trailer shall be furnished with a ball type coupler. Towbar, coupling device, and safety chains shall be provided and shall conform to ABYC and BIA standards.

3.7.2.11 Frame. Frame shall be a heavy-duty type of the manufacturer's standard. Frame shall be capable of withstanding all design loadings with no evidence of malfunction or deformation.

3.7.2.12 Fenders. Vehicle shall have fenders over wheels. The fenders shall have inside shields and shall have sufficient crown to control mud and water from splashing on chassis and boat.

3.7.2.13 Trailer bed tilting mechanism. The trailer shall be designed to allow the trailer bed to tilt down while remaining hitched to the truck to allow easier onloading and offloading of the boat in the water. The mechanism to effect this shall be of the manufacturer's standard for the load rating of the trailer.

3.7.2.14 Winch. The trailer shall be equipped with an electric, 12 volt dc powered winch with steel cable which shall be rated for loads including 6,000 lb. The winch shall have a manual override which shall be equipped with a hand crank.

3.7.2.15 Hold-down straps. Quick release nylon web safety straps for holding boat trailer in transit shall be provided.

3.7.3 Towing vehicle equipment. Towing vehicle 2-inch coupling ball and electrical receptacle and wiring harness shall also be provided to allow the system to be connected to an available Navy pick-up truck or other towing vehicle.

3.8 System construction. The system shall be designed and constructed to facilitate field maintenance. All adjustments and replaceable accessories shall be readily accessible. Conditions which can be hazardous to personnel or deleterious to equipment shall not be permitted.

3.9 Dissimilar metals. Intimate contact which can be expected to cause galvanic corrosion shall be avoided. When such contact cannot be avoided, an interposing insulating material shall be provided to minimize the corrosive effect.

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3.10 Lubrication. Lubrication shall be in accordance with the equipment manufacturer's recommendations and shall be compatible with lubricants specified herein. Pressure lubrication shall not damage seals or other parts. All parts requiring lubrication shall be lubricated prior to delivery and tagged to show the type and temperature rating of the lubricant used.

3.11 Servicing and adjusting. Prior to acceptance of the utility boat system by the Government inspector, contractor shall service and adjust the system components for operational use including at least the following: inflation of all tires, lubrication of all required boat, engine, trailer chassis, and running gear locations with grades of lubricant recommended for the ambient air temperature at the delivery point (see 3.6.18 and 3.10); adjustment and tuning of engine; adjustment of all cables and controls and mounting of all components and accessories.

3.12 Corrosion protection.

3.12.1 Trailer structural metal. All trailer structural metal parts shall be fabricated of metal having been protected by corrosion retarding processes other than paint, which are certified as protective in a salt water neighboring environment.

3.12.2 Metal boat fittings. All metal boat fittings shall be either manganese, bronze, monel, marine aluminum, or stainless steel (300 series) and shall either be inherently corrosion resistant or shall be treated so so as to retard corrosion and shall conform to ABYC standards.

3.12.3 Engine metal surfaces. Engine metal surfaces shall be protected from salt water corrosive environment by the engine manufacturer's standard method of protection.

3.12.4 Fasteners. All metal fasteners which might be exposed to corrosive environments shall be nickel-plated brass or stainless steel (300 series).

3.13 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. Interior and exterior fiberglass surfaces of the boat shall be finished using the manufacturer's standard impregnated gel coat. Color of the finish coat for both paint and gel coat shall be light grey unless otherwise specified (see 6.2).

3.13.1 Cleaning. Metal surfaces to be painted shall be cleaned to insure that they are free from all oil, grease, welding slag and spatter, mill scale which can be removed by power wire brushing, products of corrosion, dirt, or foreign substances.

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3.13.2 Treatment. As soon as practicable after cleaning, and before any evidence of rust or other contamination can result, metal surfaces shall be treated with a primer pretreatment coating to increase the adhesion of the coating system.

3.13.3 Painting. For metal surfaces not otherwise protected (see 3.12), painting shall consist of at least one coat of rust-inhibiting primer and one coat of finish enamal. The rust-inhibiting primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting, when utilized shall be with manufacturer's current materials according to manufacturer's current processes except that the total dry film thickness shall be not less than 2.5 mils. The paint shall be free from holidays, runs, sags, orange peel, or other defects.

3.14 Fungus and moisture resistance. The electrical circuitry, including all components and connections except as specified below sha be protected from the effects of fungus growth and moisture by an over treatment with a varnish conforming to MIL-V-173, except that 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish shall be used in lieu of salicylanilide, the specified fungi toxicant:

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose fuctions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.15 Marking, identification plates, instruction plates, and boat registry numbers.

3.15.1 Marking. The utility boat system shall be marked in accordance with MIL-STD-130 unless otherwise required by the contracting officer (see 6.2).

3.15.2 Identification plates. Identification data plates shall be furnished by the contracting officer, and permanently affixed in a conspicuous place on each utility boat system. Nomenclature shall be UTILITY BOAT SYSTEM, (component name). Contractor shall stamp the necessary information and affix the plate. A plate shall be included for the trailer, boat, and outboard engine components of each system.

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3.15.3 Instruction plates. The utility boat system components and accessory equipment shall be equipped with instruction plates describing the starting procedure and other special or important procedures, cautions, and warnings necessary to operate and service the components and accessories. Instruction plates shall be located for optimum visibility to the operator. Instruction marking may be included on identification plates specified in 3.15.2.

3.15.4 Boat registry numbers. Boat registry numbers, to be assigned after contract award, when required, shall be 3-inch high, black, painted numbers. The numbers shall be located as required to obtain the required certificate of numbers from the Coast Guard or the State of Registration.

3.16 Workmanship. All parts, components, and assemblies of the utility boat system including castings, forgings, molded parts, stamping, bearings, seals, and machined surfaces shall be clean and free from sand, dirt, fins, pits, sprues, scale, and other harmful extraneous material. External surfaces shall be free from burrs, dents, chipped paint, and rough or sharp edges, and corners. Fasteners shall be tight without distorting the components being held. Operable components shall function without binding or causing interference of movement.

3.16.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.16.2 Bolted connections. All bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided. All nuts, bolts, and screws shall be tight.

3.16.3 Riveted connections. All rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall completely fill the hole. Rivet heads shall be of an approved shape and shall be concentric with the rivet. Rivet head shall make full contact with the intended surface. The rivets shall be so fabricated as to develop a joint strength not less than the design value.

3.16.4 Welding. Surfaces to be welded shall be free from foreign matter which would be injurious to the weld. Welding procedures shall be in accordance with a nationally recognized code. Welds shall be of sufficient size and shape to develop the full design strength of the parts connected by the welds. Welds shall transmit imposed stresses without permanent deformation or failure when subjected to proof or service loadings.

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3.16.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Preparation for delivery inspection (see 4.7)

4.3 First article inspection. First article inspection shall be performed on one utility boat system when a first article sample is required (see 3.2). This inspection shall include the examination of 4.5 and the tests of 4.6. The first article may be a standard production item from the contractor's current inventory provided the utility boat system meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining utility boat system to be furnished under the contract.

4.4 Quality conformance inspection. Unless otherwise specified (see 6.2) quality conformance inspection shall be performed on each utility boat system. This inspection shall include the examination of 4.5.

4.5 Examination. Each utility boat system shall be examined for compliance with the requirements specified in this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

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4.6 Tests.

4.6.1 General. A compliance check for presence of all accessories specified, and compatability with stated required certifications shall be made.

4.6.2 Floor plan. A check for compliance with specified interior floor plan configuration shall be made.

4.6.3 Mechanisms and accessories. All mechanisms, mechanical and electrical, shall be operated to determine acceptability of smooth operation. Accessories mounted on brackets shall be removed and relocated on the brackets to determine acceptability of mounting configuration.

4.6.4 Hoisting. The boat and engine, while loaded with subject accessories and fully filled fuel tanks, shall be mechanically hoisted into the water, off the trailer, from a pier at least 10 feet above the water surface. Hoisting fittings and the mounting areas immediately adjacent to the fittings shall be visually inspected, after boat has been launched and retrieved, for signs of strain of hardware and/or fiberglass cracking. Damage to these items shall constitute grounds for failure of the test.

4.6.5 Launching and retrieving. The boat and engine, while fully loaded with accessories and filled fuel tanks, shall be launched from the trailer when driven onto a launching ramp. The equipped boat shall then be retrieved onto the trailer using the electric powered winch. The boat will subsequently be retested using the hand override winch system. In both cases the trailer bed tiltdown mechanism shall be utilized. All launching and retrieving equipment shall perform easily and speedily. Permanent deformation, excessive binding, (or damage to any portion of the system) will constitute failure of this test. This test will determine operational acceptability of the trailer launching assembly.

4.6.6. Towing device. The contractor shall demonstrate, to the Government representatives, the towing device's capacity to sustain six times the rated thrust of the boat-engine-propellor combination for 5 minutes, without causing visible damage to the hull structure in the areas around the connections to the hull or to any other area of the towing device. Device shall be inspected to insure tow ropes do not interfere with engine.

4.6.7 Road test. The trailer with rated payload shall be road tested for a distance of not less than 10 miles to determine conformance to the requirements of this specification.

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4.6.8 Storage, hookup, and manual manuvering. Horizontal storage, short distance manual manuvering, and hookup shall be demonstrated to determine conformance to the requirements of this specification. Hookup shall include electrical and mechanical coupling of the trailer and towing vehicle.

4.6.9 Operating test. The boat shall be deployed and in water compliance checked for mechanism, control, gage, and engine operation. Included in this test shall be seaworthyness, handling of the boat, and a static thrust test of the engine/propellor combination.

4.7 Packaging inspection. The preservation-packaging, packing, and marking of the boat system shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Preservation-packaging. The preservation-packaging shall be level A, or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Disassembly. Disassembly shall be the minimum necessary to safeguard parts known to be subject to damage or loss, and to accomplish reduction in cube. Bolts, nuts, screws, pins, and washers removed shall be reinstalled in one of the mating parts and secured to prevent their loss.

5.1.1.2 Preservatives. Preservatives shall conform to the applicable specification listed in, and shall be applied in accordance with MIL-P-116.

5.1.1.3 Unpainted and uncoated surfaces. Unpainted and uncoated exterior ferrous metal surface of the boat, trailer, and accessories shall be coated with type P-1 preservative.

5.1.1.4 Outboard motor. With the engine running smoothly at approximatedly 1,500 rpm, P-10 preservative shall be sprayed into the carburetor throat(s) until the engine chokes to a stop. A tag conforming to UU-T-81, type B, class 2 shall be attached in a conspicuous location indicating that "ENGINE PRESERVATIVE; DO NOT CRANK OR RUN". The carburetors and cooling system shall be drained dry.

5.1.1.5 Fuel tanks. The fuel tanks shall be drained of all fuel and dried. The interior of the tanks shall be coated with type P-10 preservative in a manner to insure thorough coating of all surfaces. The tanks shall be secured to the floor with the straps provided. The detached fuel lines shall be packaged in a fiberboard box conforming to PPP-B-636, class weather-resistant and cushioned to prevent movement.

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5.1.1.6 Battery. The battery shall be placed inside the battery case and cushioned to prevent movement. The battery case with cover installed shall be secured to the boat floor with the straps provided.

5.1.1.7 Life preservers. The life preservers shall be packaged in fiberboard boxes conforming to PPP-B-636, class weather-resistant and cushioned to prevent movement.

5.1.1.8 Wood paddles, boat hook, fenders, and anchor. The paddles, boat hook, fenders, and anchor shall be secured in a protected location inside the boat in a manner to prevent movement and damage during shipment.

5.1.1.9 Rope. The rope shall be coiled or hanked and secured with cotton cord or twine.

5.1.1.10 Fire extinguisher. The fire extinguisher shall be placed and secured in the mounting brackets.

5.1.1.11 First aid kit. The first aid kit shall be packaged in accordance with MIL-P-116 method IC-2.

5.1.1.12 Emergency flare kit. The kit shall be packaged in accordance with MIL-P-116 method III, in a fiberboard box conforming to PPP-B-636, class weather-resistant.

5.1.1.13 Spot lights and running lights. Spot lights and running lights, if removed, shall be packaged as specified for the emergency flare kit in 5.1.1.12.

5.1.1.14 Repair parts and maintenance tools. The preservation application criteria and applicable methods of preservation of MIL-P-116 shall be used to preserve repair parts. Unprotected surfaces of maintenance tools shall be coated with type P-1 or P-10 preservative.

5.1.1.15 Technical publications. Technical publications, when furnished, shall be preserved in accordance with MIL-P-116, method IC-3.

5.1.1.16 Consolidated packaging.

5.1.1.16.1 Repair parts, maintenance tools, technical publications, and small components. Repair parts, tools, technical publications, and small components shall be packaged in close-fitting boxes conforming to PPP-B-601, domestic type or PPP-B-636, class weather-resistant. The contents shall be cushioned, blocked, and braced to prevent movement within the boxes.

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5.1.1.16.2 Outboard motor. The outboard motor shall be packaged in a close-fitting box conforming to PPP-B-601, domestic type. The motor shall be cushioned, blocked, and braced to prevent movement within the box.

5.1.2 Level C. The complete boat system shall be preserved and packaged in a manner that will insure protection against deterioration and damage during shipment. This level may conform to the contractor's standard practice when such meets the requirements of this level.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. The safety equipment, disassembled components, and consolidated packaged components shall be placed inside the boat in a protected location, and blocked and braced to prevent movement during shipment and damage to the boat. Components in wooden boxes shall be secured to the base of the crate or placed and secured on the deck of the boat. The canvas cover shall be installed in place in a manner to prevent water pockets. The boat shall be placed on the trailer, secured with the means provided, and packed in a crate conforming to MIL-C-104, type II, class optional, style A. Cushioning, anchoring, blocking and bracing shall be in accordance with MIL-STD-1186, and the appendix to the crate specification.

5.2.2 Level B. Each complete boat system shall be packed as specified for level A, except the crate shall conform to MIL-C-3774, type II.

5.2.3 Level C. The complete boat system shall be packed for shipment in a manner which will insure arrival at destination in satisfactory condition and which will be acceptable to the carrier at lowest rates. Packing shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. Interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The utility boat system is intended to be used to tow oil spill booms into place. The system is designed to be towed to the site by an available towing vehicle, and the boat launched either from the trailer on a boat launching ramp or by the dock crane.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) When first article is required for inspection and approval (see 3.2, 4.3 and 6.4).

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(c) When outboard mounting well shroud is required (see 3.5).
(d) When the console shall be positioned other than on the starboard side of the boat (see 3.5.2.1).
(e) When water temperature indication is required (see 3.6.6).
(f) When ammeter is required (see 3.6.7).
(g) When engine log meter is required (see 3.6.8).
(h) Whether propeller shall have 3 or 4 blades and whether it shall be fabricated of manganese bronze or heat-treated, semipermanent or permanent mold aluminum-alloy casting (see 3.6.13).
(i) When gasoline other than 92 octane shall be used (see 3.6.17.1).
(j) When spare wheel assembly is not required (see 3.7.2.6).
(k) When spare tire is not required (see 3.7.2.6.1).
(l) When finish color shall be other than light grey (see 3.13).
(m) When marking other than in accordance with MIL-STD-130 is required (see 3.15.1).
(n) When quality conformance inspection of all utility boat systems is not required (see 4.4).
(o) Level of preservation-packaging and level of packing required (see 5.1 and 5.2).

6.3 Contract data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Armed Services Procurement Regulation (ASPR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of paragraph 7-104.55 of the ASPR. The first article should be a preproduction sample or it may be a standard production item from the contractor's current inventory as specified in 4.3. The first article should consist of one unit. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, test, and approval of the first article.

Preparing activity:

Navy - YD

Project No. 1940-N039

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

1940-N039

DOCUMENT IDENTIFIER AND TITLE BOAT SYSTEM, UTILITY, NOMINAL, 20 FOOT REINFORCED PLASTIC WITH 85 HORSEPOWER OUTBOARD ENGINE AND TRAILER

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

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